Nidharsan V – 22CSR130

DEVOPS DAY 5 – Deploy Java app to minikube automated with jenkins

Deploy Java App to Minikube Automated with Jenkins

1. Overview

Automating the deployment of a Java application to Minikube using Jenkins involves building the application, creating a Docker image, pushing it to a container registry, and deploying it to Minikube using Kubernetes manifests.

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2. Key Concepts

A. Jenkins Pipeline

Jenkins automates the CI/CD process using a declarative pipeline. The pipeline consists of multiple stages such as:

- **SCM Checkout:** Fetches code from a repository (GitHub/GitLab).
- **Build & Test:** Uses Maven (mvn package) to compile and test the Java application.
- **Docker Build & Push:** Builds a Docker image of the application and pushes it to Docker Hub.
- **Deploy to Minikube:** Uses kubectl to apply Kubernetes deployment and service files.

B. Minikube

Minikube is a lightweight Kubernetes cluster for local development and testing. It allows developers to run Kubernetes locally and deploy applications without needing a cloud-based cluster.

Commands to Start Minikube:

sh

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minikube start

kubectl cluster-info

kubectl get nodes

C. Docker

Docker is used to package the Java application into a container image, making it portable and easy to deploy across environments.

Dockerfile Example:

dockerfile

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FROM openjdk:11

COPY target/webapp.jar /app/webapp.jar

WORKDIR /app

CMD ["java", "-jar", "webapp.jar"]

D. Kubernetes Deployment

Kubernetes YAML files define how the application should be deployed inside the Minikube cluster.

Deployment YAML Example:

yaml

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apiVersion: apps/v1

kind: Deployment

metadata:

  name: webapp

spec:

  replicas: 1

  selector:

```
  matchLabels:
    app: webapp
 template:
  metadata:
    labels:
      app: webapp
  spec:
    containers:
      - name: webapp
        image: nidharsan8008/war-jenkins1
        ports:
          - containerPort: 8080
```

Apply Deployment:

```
sh
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kubectl apply -f deployment.yml
kubectl get pods
```
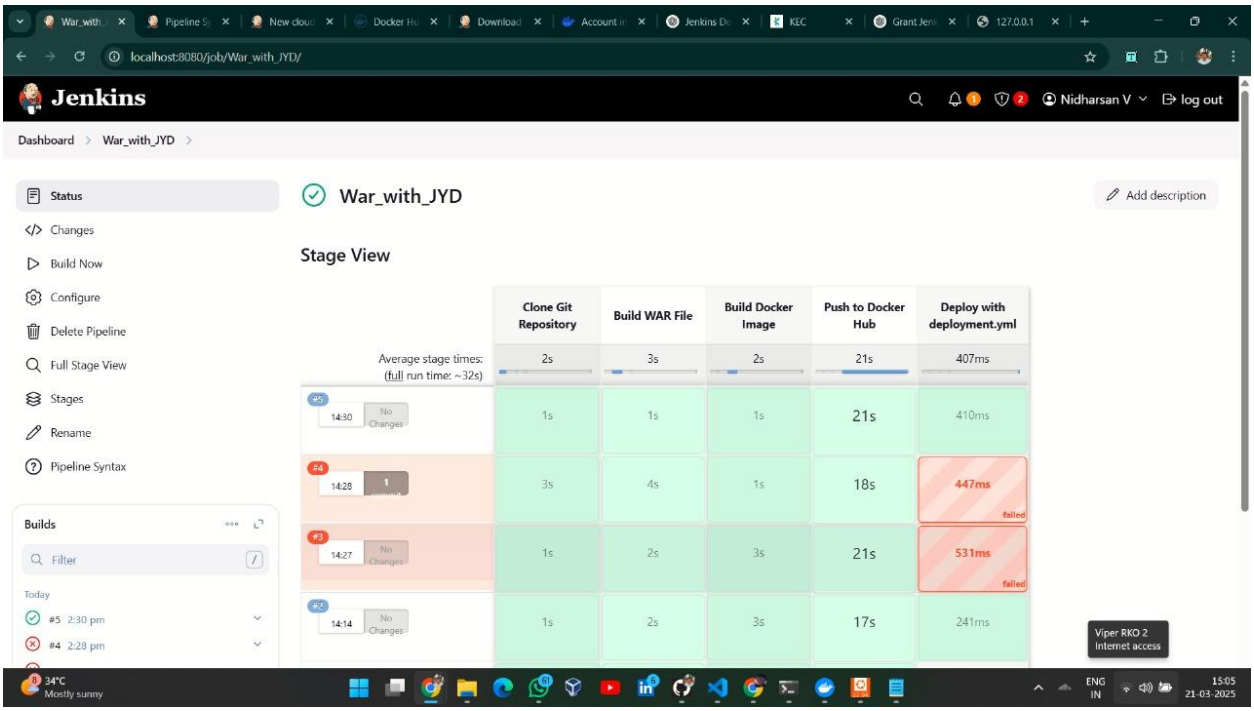
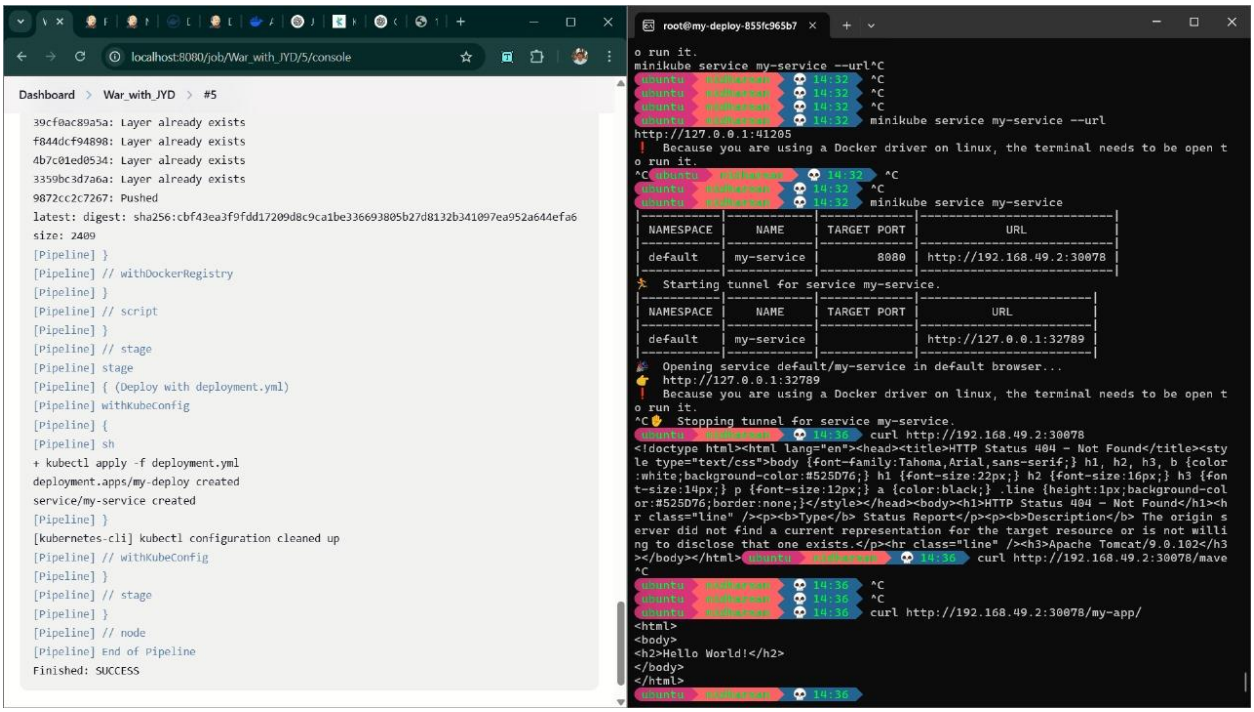

```
pipeline {
  agent any

  stages {
    stage('Clone Git Repository') {
      steps {
        git 'https://github.com/NIDHARSAN-V/Devops_Guvi_WAR.git'
```

```
    }
}

stage('Build WAR File') {
    steps {
        sh 'mvn clean package'  // Build WAR file using Maven
    }
}

stage('Build Docker Image') {
    steps {
        script {
            sh 'docker build -t warimage-jenkins .'
            sh 'docker tag warimage-jenkins nidharsan8008/warimage-jenkins1'
        }
    }
}

// stage('Run Docker Container') {  // Optional stage to test the container
//     steps {
//         script {
//             sh 'docker run -d -p 8888:8080 --name war-container-jenkins warimage-jenkins'
//         }
//     }
// }
```

```
        stage('Push to Docker Hub') {
            steps {
                script {
                    withDockerRegistry(credentialsId: 'dockerhub_token', url:
'https://index.docker.io/v1/') {
                        sh 'docker push nidharsan8008/warimage-jenkins1'
                    }
                }
            }
        }


        stage('Deploy with deployment.yml') {
            steps {
                withKubeConfig(caCertificate: '', clusterName: 'minikube ', contextName: 'minikube',
credentialsId: 'MiniKube_ID', namespace: '', restrictKubeConfigAccess: false, serverUrl:
'https://192.168.49.2:8443') {
                        sh 'kubectl apply -f deployment.yml'
                }
            }
        }


        // stage('Test') {
        //     steps {
```

```
        //        withKubeConfig(caCertificate: '', clusterName: 'minikube ', contextName: 'minikube', credentialsId: 'MiniKube_ID', namespace: '', restrictKubeConfigAccess: false, serverUrl: 'https://192.168.49.2:32274') {
        //              sh ''
        //        }
        //    }
        // }



    }
}
```


Jenkins
Nidharsan V
log out
Dashboard > War_with_JYD >
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
War_with_JYD
Add description
Stage View
Clone Git Repository
Build WAR File
Build Docker Image
Push to Docker Hub
Deploy with deployment.yml
Average stage times:
(full run time: ~32s)
2s
3s
2s
21s
407ms
#5 14:30 No Changes
1s
1s
1s
21s
410ms
#4 14:28
3s
4s
1s
18s
447ms
failed
#3 14:27 No Changes
1s
2s
3s
21s
531ms
failed
#2 14:14 No Changes
1s
2s
3s
17s
241ms
Builds
Filter
Today
#5 2:30 pm
#4 2:28 pm
Viper RKO 2
Internet access
34°C
Mostly sunny
ENG
IN
15:05
21-03-2025


localhost:8080/job/War_with_JYD/5/console
Dashboard > War_with_JYD > #5
39cf0ac89a5a: Layer already exists
f844dcf94898: Layer already exists
4b7c01ed0534: Layer already exists
3359bc3d7a6a: Layer already exists
9872cc2c7267: Pushed
latest: digest: sha256:cbf43ea3f9fdd17209d8c9ca1be336693805b27d8132b341097ea952a644efa6
size: 2409
[Pipeline] }
[Pipeline] // withDockerRegistry
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy with deployment.yml)
[Pipeline] withKubeConfig
[Pipeline] {
[Pipeline] sh
+ kubectl apply -f deployment.yml
deployment.apps/my-deploy created
service/my-service created
[Pipeline] }
[kubernetes-cli] kubectl configuration cleaned up
[Pipeline] // withKubeConfig
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
root@my-deploy-855fc965b7
o run it.
minikube service my-service --url^C
minikube service my-service --url
http://127.0.0.1:41205
Because you are using a Docker driver on linux, the terminal needs to be open t
o run it.
minikube service my-service
NAMESPACE | NAME | TARGET PORT | URL
default | my-service | 8080 | http://192.168.49.2:30078
Starting tunnel for service my-service.
NAMESPACE | NAME | TARGET PORT | URL
default | my-service | | http://127.0.0.1:32789
Opening service default/my-service in default browser...
http://127.0.0.1:32789
Because you are using a Docker driver on linux, the terminal needs to be open t
o run it.
Stopping tunnel for service my-service.
curl http://192.168.49.2:30078
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><sty
le type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color
:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {fon
t-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-col
or:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><h
r class="line" /> Type Status Report Description The origin s
erver did not find a current representation for the target resource or is not willi
ng to disclose that one exists. <hr class="line" /><h3>Apache Tomcat/9.0.102</h3
></body></html> curl http://192.168.49.2:30078/mave
curl http://192.168.49.2:30078/my-app/
<html>
<body>
<h2>Hello World!</h2>
</body>
</html>